

## Compte rendu de TP

# Git et GitHub

Gestion de versions et collaboration de code

## Du depot local au travail collaboratif

*Initialisation, index, commits, branches, historique et fusion*

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# Introduction

---

Git est aujourd'hui l'outil de reference pour suivre l'evolution d'un projet logiciel. Ce rapport retrace, etape par etape, les manipulations realisees pendant le TP Git et GitHub. L'idee directrice est simple : chaque commande laisse une trace dans le depot, et les captures fournies servent de preuve concrete de ce qui a ete execute. On y suit le parcours classique d'un developpeur, depuis l'initialisation d'un depot vide jusqu'a la resolution d'un conflit de fusion.

Les captures sont presentees dans leur ordre chronologique afin de respecter la logique d'apprentissage. Plutot que de recopier integralement chaque sortie de terminal, le rapport en degage l'intention : ce que la commande cherche a faire et ce qu'elle change reellement dans le depot.

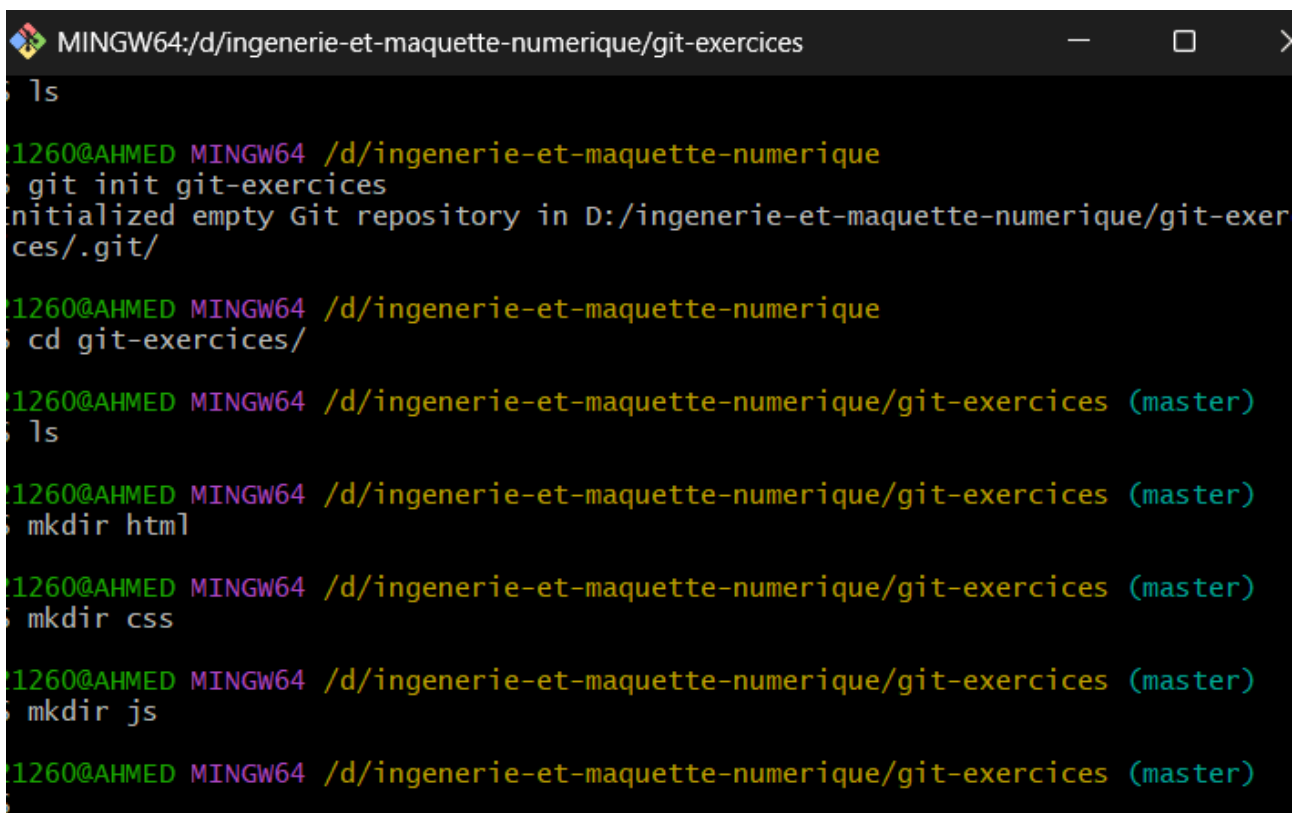
# Demarrage du depot

---

## Objectif

Tout projet Git commence par un acte fondateur : transformer un simple dossier en depot versionne. Cette premiere partie illustre cette mise en place, depuis la creation de la structure du projet jusqu'au tout premier commit qui fige une version de reference.

```
git init
git status
git add .
git commit -m "initial commit"
```



```
MINGW64:/d/ingenierie-et-maquette-numerique/git-exercices
ls
1260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique
$ git init git-exercices
initialized empty Git repository in D:/ingenierie-et-maquette-numerique/git-exercices/.git/
1260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique
$ cd git-exercices/
1260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (master)
$ ls
1260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (master)
$ mkdir html
1260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (master)
$ mkdir css
1260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (master)
$ mkdir js
1260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (master)
```

Fig. 1. – Capture 1 - Preparation du projet Git.

```
MINGW64:/d/ingenierie-et-maquette-numerique/git-exercices

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (master)
$ cd html/

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices/html (master)
$ touch index.html

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices/html (master)
$ cd ..

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (master)
$ touch css/style.css

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (master)
$ touch js/index.js

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (master)
$
```

Fig. 2. – Capture 2 - Verification de l'état du depot.

```
index.html u x
html > index.html > html:ino-js
1 </!DOCTYPE html>
2 <html class="no-js">
3 <head>
4 <meta charset="utf-8">
5 <meta http-equiv="X-UA-Compatible" content="IE=edge">
6 <title>Git exercices</title>
7 <meta name="description" content="">
8 <meta name="viewport" content="width=device-width, initial-scale=1">
9 <link rel="stylesheet" href="css/style.css">
10 </head>
11 <body>
12 <script src="js/index.js" async defer></script>
13 </body>
14 </html>
```

```
MINGW64:/d/ingenierie-et-maquette-numerique/git-exercices

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (master)
$ cd html/

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices/html (master)
$ touch index.html

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices/html (master)
$ cd ..

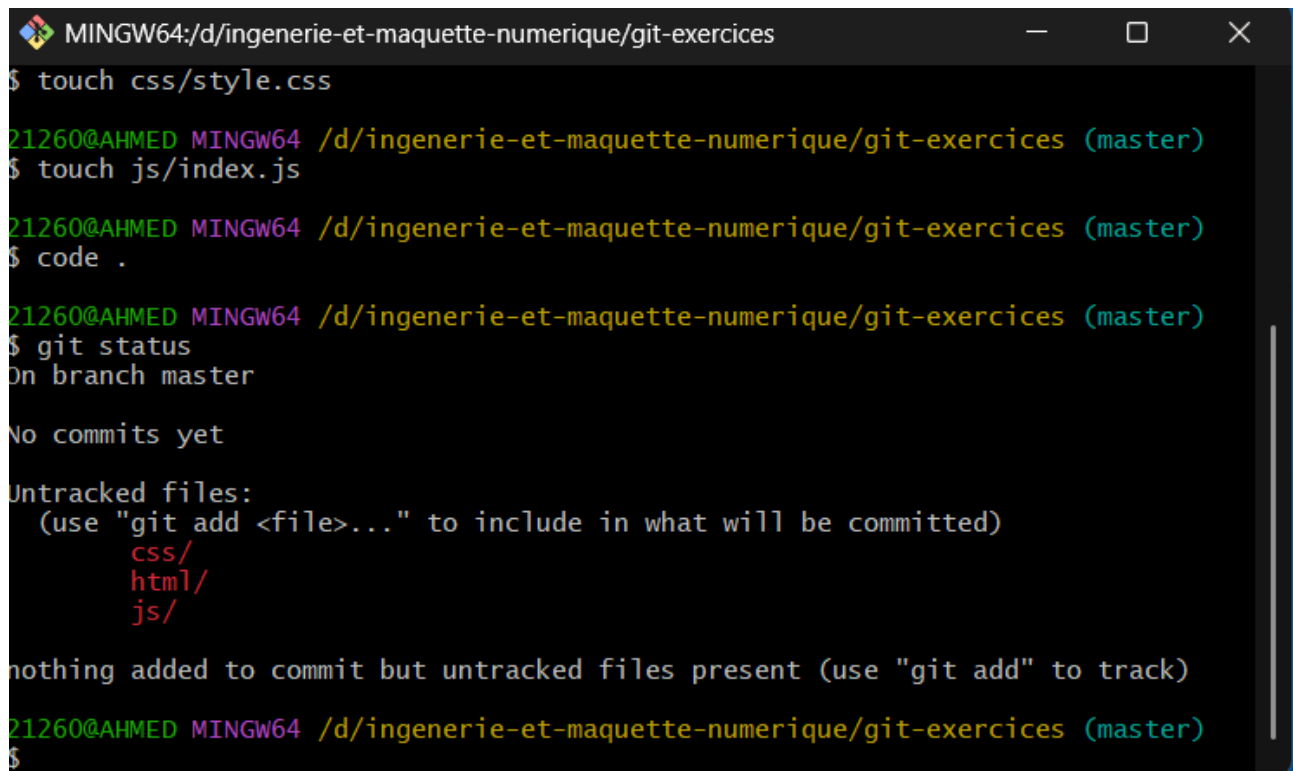
21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (master)
$ touch css/style.css

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (master)
$ touch js/index.js

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (master)
$ code .

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (master)
$
```

Fig. 3. – Capture 3 - Ajout des fichiers et premier suivi Git.

A terminal window titled 'MINGW64:/d/ingenierie-et-maquette-numerique/git-exercices' with standard window controls. The terminal shows a series of commands and their outputs. The user creates 'css/style.css' and 'js/index.js', then runs 'git status'. The output indicates no commits yet and lists 'css/', 'html/', and 'js/' as untracked files. The prompt '\$' is visible at the bottom.

```
MINGW64:/d/ingenierie-et-maquette-numerique/git-exercices
$ touch css/style.css

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (master)
$ touch js/index.js

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (master)
$ code .

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (master)
$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
      css/
      html/
      js/

nothing added to commit but untracked files present (use "git add" to track)

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (master)
$
```

Fig. 4. – Capture 4 - Premiere validation des changements.

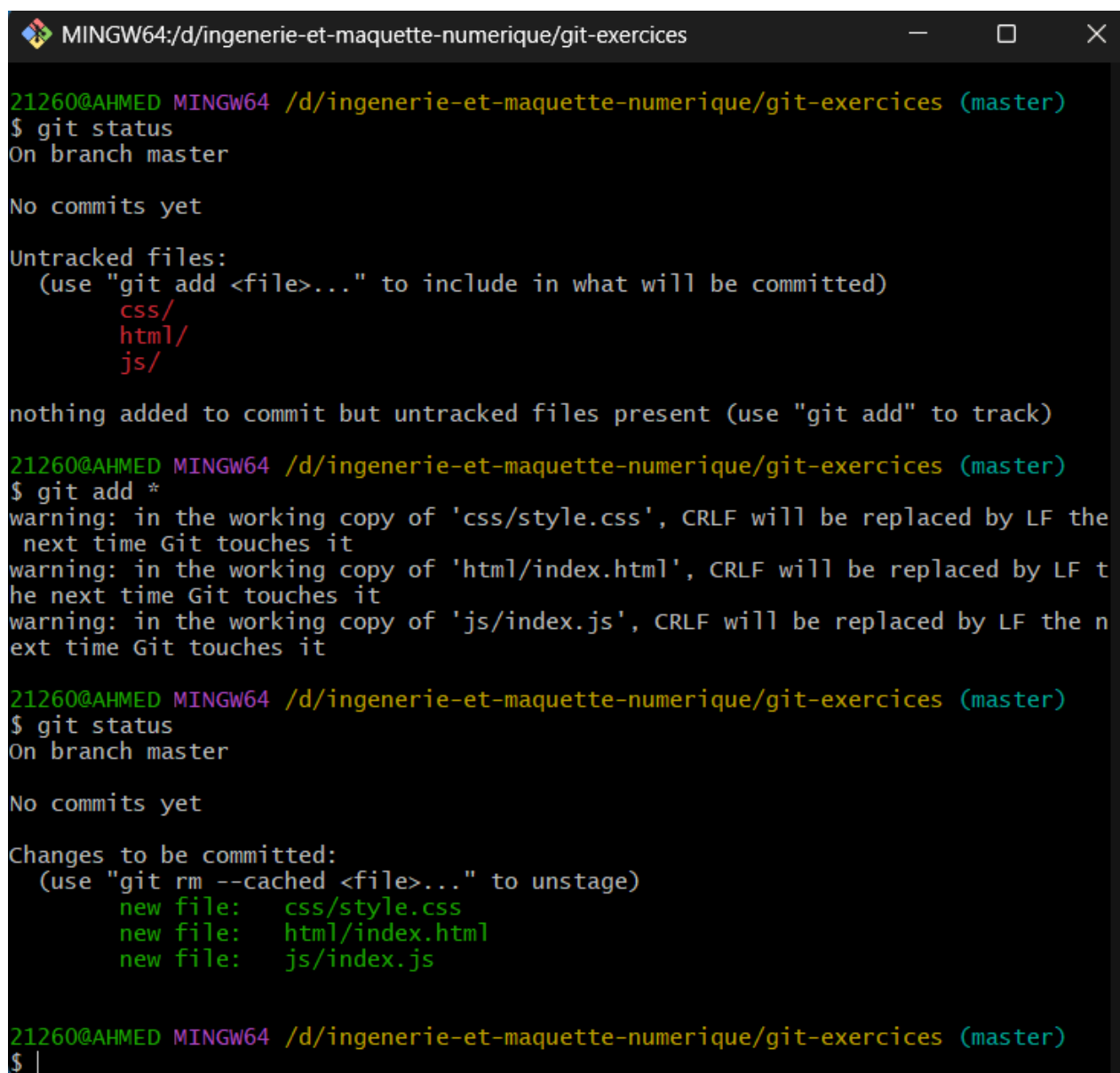
# Suivi des changements

---

## Objectif

Le coeur de Git repose sur trois espaces distincts : le repertoire de travail ou l'on edite, l'index qui prepare la prochaine validation, et l'historique qui conserve les versions confirmees. Cette section met ces notions en evidence en suivant le cycle ajout, validation et consultation.

```
git status
git add <fichier>
git commit -m "message"
git log
```



```
MINGW64:/d/ingenierie-et-maquette-numerique/git-exercices

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (master)
$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        css/
        html/
        js/

nothing added to commit but untracked files present (use "git add" to track)

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (master)
$ git add *
warning: in the working copy of 'css/style.css', CRLF will be replaced by LF the
next time Git touches it
warning: in the working copy of 'html/index.html', CRLF will be replaced by LF t
he next time Git touches it
warning: in the working copy of 'js/index.js', CRLF will be replaced by LF the n
ext time Git touches it

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (master)
$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   css/style.css
        new file:   html/index.html
        new file:   js/index.js

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (master)
$ |
```

Fig. 5. – Capture 5 - Etat du projet apres ajout des fichiers.

```
MINGW64:/d/ingenierie-et-maquette-numerique/git-exercices
nothing added to commit but untracked files present (use "git add" to trac

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (mast
$ git add *
warning: in the working copy of 'css/style.css', CRLF will be replaced by
next time Git touches it
warning: in the working copy of 'html/index.html', CRLF will be replaced b
he next time Git touches it
warning: in the working copy of 'js/index.js', CRLF will be replaced by LF
ext time Git touches it

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (mast
$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   css/style.css
    new file:   html/index.html
    new file:   js/index.js

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (mast
$ git commit -m "first commit"
[master (root-commit) 32fa2e5] first commit
 3 files changed, 18 insertions(+)
 create mode 100644 css/style.css
 create mode 100644 html/index.html
 create mode 100644 js/index.js

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (mast
$ git status
On branch master
nothing to commit, working tree clean

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (mast
$ |
```

Fig. 6. – Capture 6 - Controle de l'index Git.



```
MINGW64:/d/ingenierie-et-maquette-numerique/git-exercices
warning: in the working copy of 'js/index.js', CRLF will be replaced by LF the next time Git touches it

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (master)
$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   css/style.css
        new file:   html/index.html
        new file:   js/index.js

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (master)
$ git commit -m "first commit"
[master (root-commit) 32fa2e5] first commit
3 files changed, 18 insertions(+)
create mode 100644 css/style.css
create mode 100644 html/index.html
create mode 100644 js/index.js

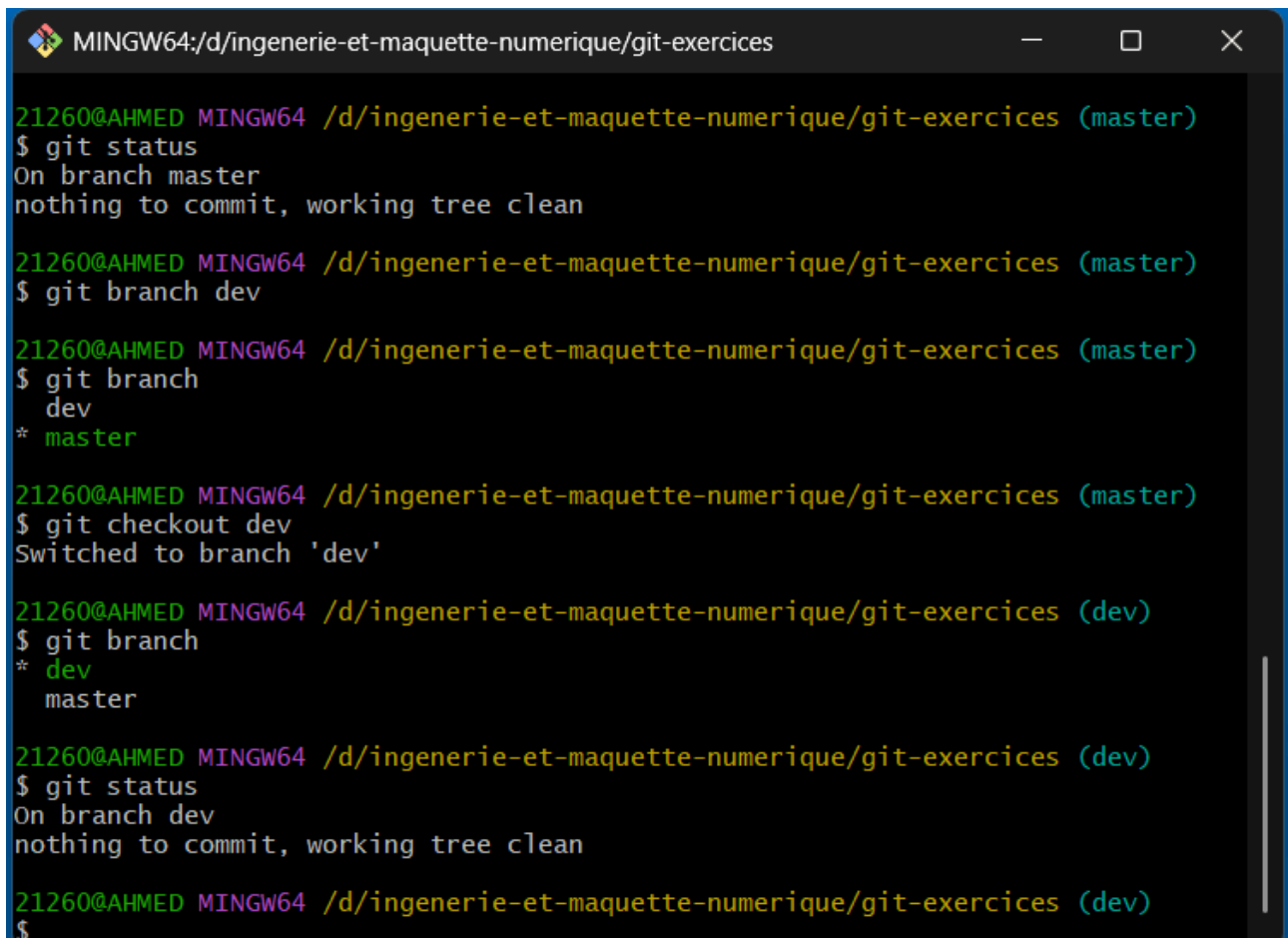
21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (master)
$ git status
On branch master
nothing to commit, working tree clean

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (master)
$ git branch dev

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (master)
$ git branch
dev
* master

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (master)
$ |
```

Fig. 7. – Capture 7 - Validation d'un changement.



```
MINGW64:/d/ingenierie-et-maquette-numerique/git-exercices
21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (master)
$ git status
On branch master
nothing to commit, working tree clean

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (master)
$ git branch dev

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (master)
$ git branch
dev
* master

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (master)
$ git checkout dev
Switched to branch 'dev'

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (dev)
$ git branch
* dev
  master

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (dev)
$ git status
On branch dev
nothing to commit, working tree clean

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (dev)
$
```

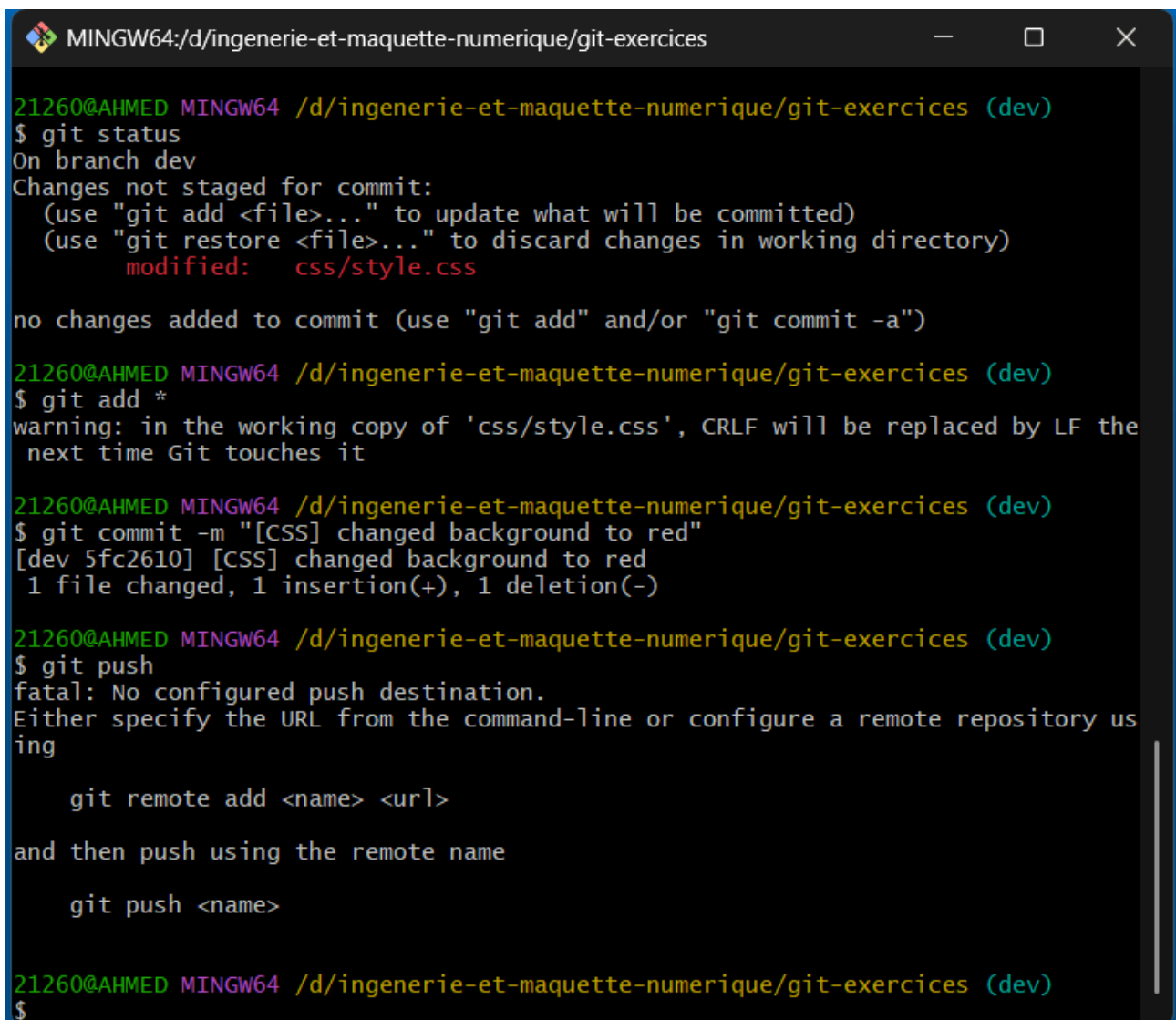
Fig. 8. – Capture 8 - Verification apres commit.

# Branches et navigation

## Objectif

Les branches permettent de développer une fonctionnalité à l'écart de la version principale, sans risque pour le code stable. Les captures suivantes montrent comment créer une branche, passer de l'une à l'autre et mesurer les écarts entre deux versions avant toute intégration.

```
git branch
git branch <nom-branche>
git checkout <nom-branche>
git diff <branche-a> <branche-b>
```



```
MINGW64:/d/ingenierie-et-maquette-numerique/git-exercices

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (dev)
$ git status
On branch dev
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   css/style.css

no changes added to commit (use "git add" and/or "git commit -a")

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (dev)
$ git add *
warning: in the working copy of 'css/style.css', CRLF will be replaced by LF the
next time Git touches it

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (dev)
$ git commit -m "[CSS] changed background to red"
[dev 5fc2610] [CSS] changed background to red
1 file changed, 1 insertion(+), 1 deletion(-)

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (dev)
$ git push
fatal: No configured push destination.
Either specify the URL from the command-line or configure a remote repository us
ing

    git remote add <name> <url>

and then push using the remote name

    git push <name>

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (dev)
$
```

Fig. 9. – Capture 9 - Liste ou création de branches.

```
21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (dev)
$ git checkout master
Switched to branch 'master'

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (master)
$ cat css/style.css
body {
  background: blue;
}

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (master)
$ git pull
There is no tracking information for the current branch.
Please specify which branch you want to merge with.
See git-pull(1) for details.

    git pull <remote> <branch>

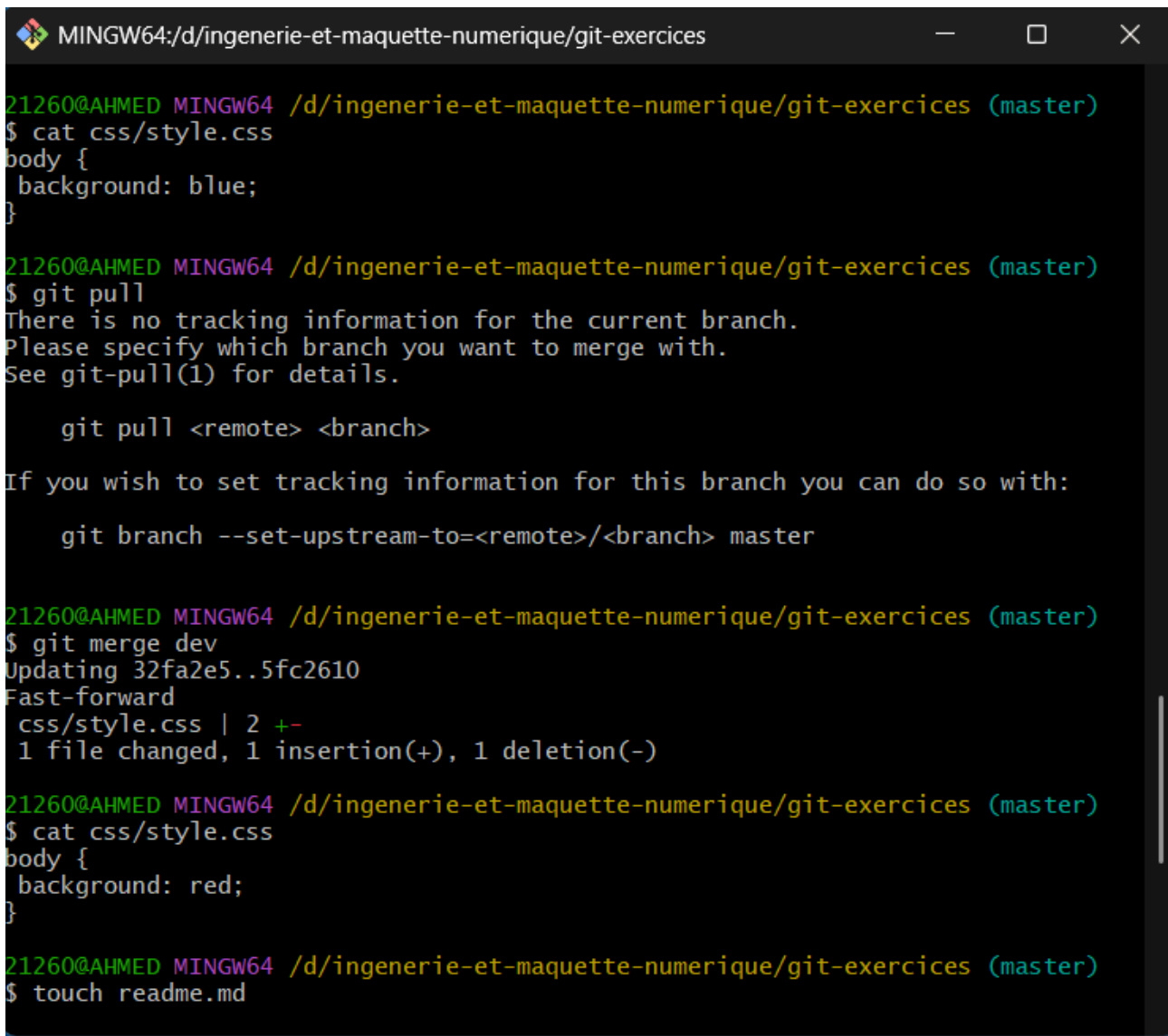
If you wish to set tracking information for this branch you can do so with:

    git branch --set-upstream-to=<remote>/<branch> master

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (master)
$ git merge dev
Updating 32fa2e5..5fc2610
Fast-forward
 css/style.css | 2 +-
 1 file changed, 1 insertion(+), 1 deletion(-)

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (master)
$ |
```

Fig. 10. – Capture 10 - Changement de branche.



```
MINGW64:/d/ingenierie-et-maquette-numerique/git-exercices
21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (master)
$ cat css/style.css
body {
  background: blue;
}

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (master)
$ git pull
There is no tracking information for the current branch.
Please specify which branch you want to merge with.
See git-pull(1) for details.

    git pull <remote> <branch>

If you wish to set tracking information for this branch you can do so with:

    git branch --set-upstream-to=<remote>/<branch> master

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (master)
$ git merge dev
Updating 32fa2e5..5fc2610
Fast-forward
 css/style.css | 2 +-
 1 file changed, 1 insertion(+), 1 deletion(-)

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (master)
$ cat css/style.css
body {
  background: red;
}

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (master)
$ touch readme.md
```

Fig. 11. – Capture 11 - Comparaison entre branches.

```
MINGW64:/d/ingenierie-et-maquette-numerique/git-exercices
background: red;
}

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (master)
$ touch readme.md

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (master)
$ git status
On branch master
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        readme.md

nothing added to commit but untracked files present (use "git add" to track)

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (master)
$ git checkout dev
Switched to branch 'dev'

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (dev)
$ git add readme.md

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (dev)
$ git commit -m "Added a readme file"
[dev 8f4d75c] Added a readme file
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 readme.md

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (dev)
$ git diff master dev
diff --git a/readme.md b/readme.md
new file mode 100644
index 0000000..e69de29

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (dev)
$ cat css/style.css
body {
background: blue;
}
```

Fig. 12. – Capture 12 - Controle de l'état apres navigation.

```
MINGW64:/d/ingenierie-et-maquette-numerique/git-exercices
21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (d
$ git branch branch-a

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (d
ev)
$ git branch branch-b

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (d
ev)
$ git diff branch-a branch-b

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (d
ev)
$ git log branch-a
commit 8f4d75ccd0775f683887ad470229d905d5eb495a (HEAD -> dev, branch-b,
branch-a)
Author: Ahmed BenAhmed <ahmed.tana1t666@gmail.com>
Date: Wed Feb 18 09:32:53 2026 +0000

    Added a readme file

commit 5fc26106d855790d24f686ee6da68887e79438c5 (master)
Author: Ahmed BenAhmed <ahmed.tana1t666@gmail.com>
Date: Wed Feb 18 09:28:57 2026 +0000

    [CSS] changed background to red

commit 32fa2e5edd967482dbda8b16ab55e3662aea786e
Author: Ahmed BenAhmed <ahmed.tana1t666@gmail.com>
Date: Wed Feb 18 09:25:41 2026 +0000

    first commit

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (d
ev)
$ |
```

Fig. 13. – Capture 13 - Observation des differences.

```
MINGW64:/d/ingenierie-et-maquette-numerique/git-exercices
[CSS] changed background to red

commit 32fa2e5edd967482dbda8b16ab55e3662aea786e
Author: Ahmed BenAhmed <ahmed.tanalt666@gmail.com>
Date:   Wed Feb 18 09:25:41 2026 +0000

    first commit

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (dev)
$ git checkout branch-a
Switched to branch 'branch-a'

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (branch-a)
$ nano readme.md

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (branch-a)
$ cat readme.md
Branch-a instructions...

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (branch-a)
$ git add readme.md

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (branch-a)
$ git commit -m "[ReadMe] Added branch a instructions"
[branch-a e42f47d] [ReadMe] Added branch a instructions
1 file changed, 2 insertions(+)

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (branch-a)
$ |
```

Fig. 14. – Capture 14 - Suivi de l'historique de branche.

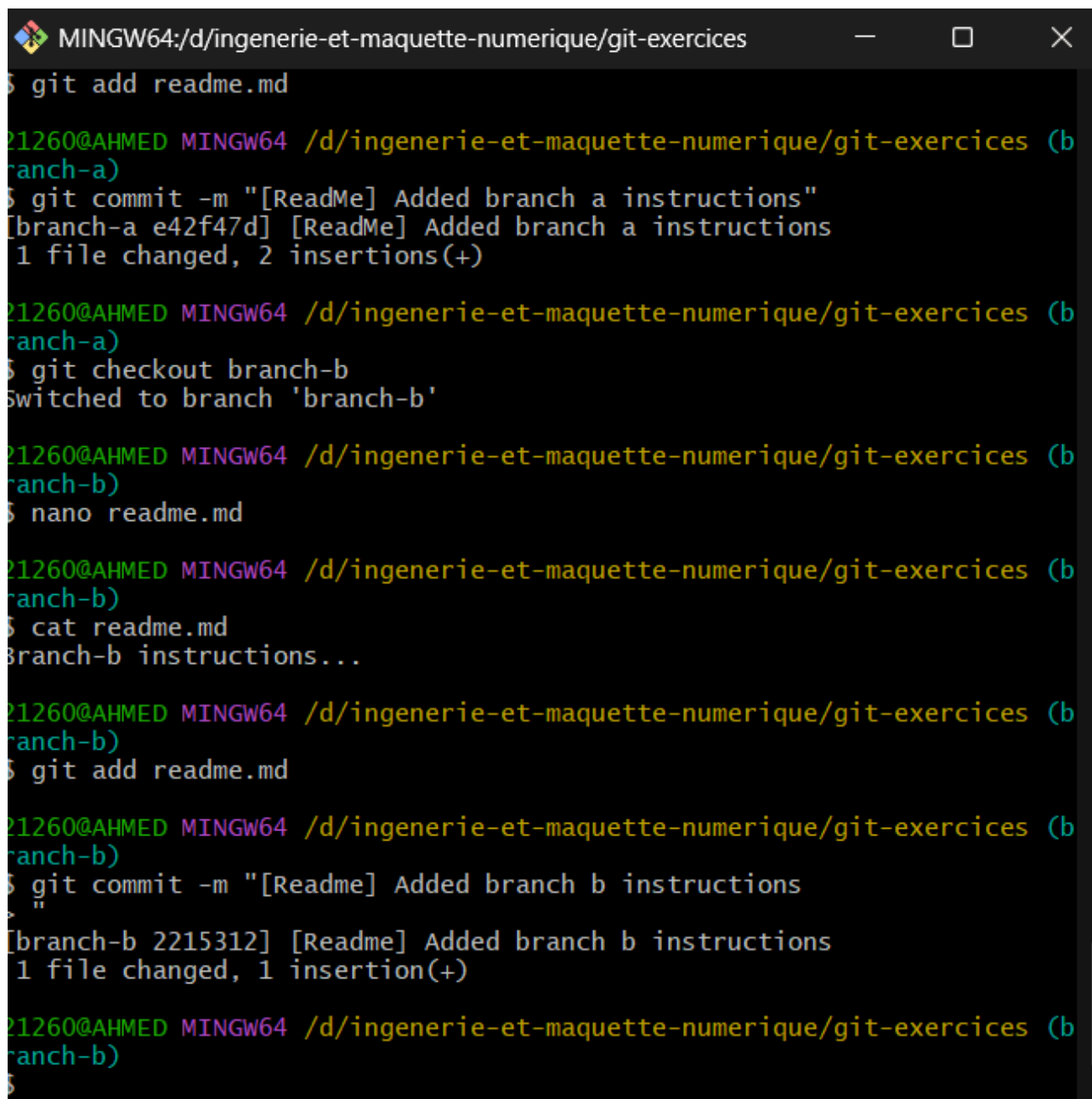


## Modifications avancees

### Objectif

Au-dela des commandes de base, Git offre des outils pour reprendre la main sur le contenu du depot : retirer un fichier du suivi, restaurer une version anterieure ou relire l'historique sous une forme condensee. Cette section documente ces operations de controle.

```
git rm <fichier>
git restore <fichier>
git reset <revision>
git log --oneline --graph
```



```
MINGW64:/d/ingenierie-et-maquette-numerique/git-exercices
$ git add readme.md

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (branch-a)
$ git commit -m "[ReadMe] Added branch a instructions"
[branch-a e42f47d] [ReadMe] Added branch a instructions
1 file changed, 2 insertions(+)

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (branch-a)
$ git checkout branch-b
Switched to branch 'branch-b'

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (branch-b)
$ nano readme.md

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (branch-b)
$ cat readme.md
Branch-b instructions...

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (branch-b)
$ git add readme.md

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (branch-b)
$ git commit -m "[Readme] Added branch b instructions"
[branch-b 2215312] [Readme] Added branch b instructions
1 file changed, 1 insertion(+)
```

Fig. 15. – Capture 15 - Manipulation avancee dans le depot.

```
MINGW64:/d/ingenierie-et-maquette-numerique/git-exercices
commit 32fa2e5edd967482dbda8b16ab55e3662aea786e
Author: Ahmed BenAhmed <ahmed.tana1t666@gmail.com>
Date:   Wed Feb 18 09:25:41 2026 +0000

    first commit

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (branch-b)
$ git log branch-b
commit 221531229a3f9a8f3047d0bf77230797f9a2ac96 (HEAD -> branch-b)
Author: Ahmed BenAhmed <ahmed.tana1t666@gmail.com>
Date:   Wed Feb 18 09:41:53 2026 +0000

    [Readme] Added branch b instructions

commit 8f4d75ccd0775f683887ad470229d905d5eb495a (dev)
Author: Ahmed BenAhmed <ahmed.tana1t666@gmail.com>
Date:   Wed Feb 18 09:32:53 2026 +0000

    Added a readme file

commit 5fc26106d855790d24f686ee6da68887e79438c5 (master)
Author: Ahmed BenAhmed <ahmed.tana1t666@gmail.com>
Date:   Wed Feb 18 09:28:57 2026 +0000

    [CSS] changed background to red

commit 32fa2e5edd967482dbda8b16ab55e3662aea786e
Author: Ahmed BenAhmed <ahmed.tana1t666@gmail.com>
Date:   Wed Feb 18 09:25:41 2026 +0000

    first commit

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (branch-b)
$ |
```

Fig. 16. – Capture 16 - Verification de l'espace de travail.

```
MINGW64:/d/ingenierie-et-maquette-numerique/git-exercices
21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (branch-master)
$ git merge branch-a
Updating 8f4d75c..e42f47d
Fast-forward
 README.md | 2 ++
 1 file changed, 2 insertions(+)

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (branch-master)
$ git diff branch-a branch-master

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (branch-master)
$ git merge branch-b
Auto-merging README.md
CONFLICT (content): Merge conflict in README.md
Automatic merge failed; fix conflicts and then commit the result.

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (branch-master|MERGING)
$ git status
On branch branch-master
You have unmerged paths.
  (fix conflicts and run "git commit")
  (use "git merge --abort" to abort the merge)

Unmerged paths:
  (use "git add <file>..." to mark resolution)
        both modified:   README.md

no changes added to commit (use "git add" and/or "git commit -a")

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (branch-master|MERGING)
$
```

Fig. 17. – Capture 17 - Consultation de l'historique.

```
21260@AHMED MINGW64 /d/ingenerie-et-maquette-numerique/git-exercices (branch-master)
$ git log
commit c3e04f358555484619e14abe1e8bdfb8b1802c93 (HEAD -> branch-master)
Merge: e42f47d 2215312
Author: Ahmed BenAhmed <ahmed.tanalt666@gmail.com>
Date: Wed Feb 18 09:47:25 2026 +0000

    making peace between branch-a and branch-b

commit 221531229a3f9a8f3047d0bf77230797f9a2ac96 (branch-b)
Author: Ahmed BenAhmed <ahmed.tanalt666@gmail.com>
Date: Wed Feb 18 09:41:53 2026 +0000

    [Readme] Added branch b instructions

commit e42f47dec9f47d316090e089405b75b71fde053d (branch-a)
Author: Ahmed BenAhmed <ahmed.tanalt666@gmail.com>
Date: Wed Feb 18 09:40:20 2026 +0000

    [ReadMe] Added branch a instructions

commit 8f4d75ccd0775f683887ad470229d905d5eb495a (dev)
Author: Ahmed BenAhmed <ahmed.tanalt666@gmail.com>
Date: Wed Feb 18 09:32:53 2026 +0000

    Added a readme file

commit 5fc26106d855790d24f686ee6da68887e79438c5 (master)
Author: Ahmed BenAhmed <ahmed.tanalt666@gmail.com>
Date: Wed Feb 18 09:28:57 2026 +0000

    [CSS] changed background to red

commit 32fa2e5edd967482dbda8b16ab55e3662aea786e
Author: Ahmed BenAhmed <ahmed.tanalt666@gmail.com>
Date: Wed Feb 18 09:25:41 2026 +0000
```

Fig. 18. – Capture 18 - Etat apres modifications.

```
1260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (branch-master)
$ git diff branch-master dev
diff --git a/readme.md b/readme.md
index 94eaa2c..e69de29 100644
--- a/readme.md
+++ b/readme.md
@@ -1,2 +0,0 @@
-Branch-a instructions...
-Branch-b instructions...

1260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (branch-master)
$ git diff dev branch-master
diff --git a/readme.md b/readme.md
index e69de29..94eaa2c 100644
--- a/readme.md
+++ b/readme.md
@@ -0,0 +1,2 @@
+Branch-a instructions...
+Branch-b instructions...
```

Fig. 19. – Capture 19 - Controle des fichiers suivis.

```
MINGW64:/d/ingenierie-et-maquette-numerique/git-exercices

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (branch-master)
$ git checkout master
Switched to branch 'master'

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (master)
$ git merge branch-master
Updating 5fc2610..c3e04f3
Fast-forward
 README.md | 2 ++
 1 file changed, 2 insertions(+)
 create mode 100644 README.md

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (master)
$ git log master
commit c3e04f358555484619e14abe1e8bdfb8b1802c93 (HEAD -> master, branch-master)
Merge: e42f47d 2215312
Author: Ahmed BenAhmed <ahmed.tanalt666@gmail.com>
Date: Wed Feb 18 09:47:25 2026 +0000

    making peace between branch-a and branch-b

commit 221531229a3f9a8f3047d0bf77230797f9a2ac96 (branch-b)
Author: Ahmed BenAhmed <ahmed.tanalt666@gmail.com>
Date: Wed Feb 18 09:41:53 2026 +0000

    [Readme] Added branch b instructions

commit e42f47dec9f47d316090e089405b75b71fde053d (branch-a)
Author: Ahmed BenAhmed <ahmed.tanalt666@gmail.com>
Date: Wed Feb 18 09:40:20 2026 +0000

    [ReadMe] Added branch a instructions

commit 8f4d75ccd0775f683887ad470229d905d5eb495a (dev)
Author: Ahmed BenAhmed <ahmed.tanalt666@gmail.com>
Date: Wed Feb 18 09:32:53 2026 +0000
```

Fig. 20. – Capture 20 - Inspection des commits.

```
MINGW64:/d/ingenierie-et-maquette-numerique/git-exercices
Added a readme file

commit 5fc26106d855790d24f686ee6da68887e79438c5
Author: Ahmed BenAhmed <ahmed.tanalt666@gmail.com>
Date:   Wed Feb 18 09:28:57 2026 +0000

    [CSS] changed background to red

commit 32fa2e5edd967482dbda8b16ab55e3662aea786e
Author: Ahmed BenAhmed <ahmed.tanalt666@gmail.com>
Date:   Wed Feb 18 09:25:41 2026 +0000

    first commit

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (master)
$

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (master)
$ git branch -d dev
Deleted branch dev (was 8f4d75c).

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (master)
$ git branch -d branch-a
Deleted branch branch-a (was e42f47d).

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (master)
$ git branch -d branch-b
Deleted branch branch-b (was 2215312).

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (master)
$ git branch -d branch-master
Deleted branch branch-master (was c3e04f3).

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices (master)
$
```

Fig. 21. – Capture 21 - Verification finale de cette etape.

## Fusion et finalisation

---

### Objectif

Fusionner, c'est reunir le travail mene sur plusieurs branches en une seule ligne d'historique. Cette partie regroupe les operations de cloture du TP : integration des branches, derniers controles de l'etat du depot et lecture de l'historique global.

```
git merge <branche>
git status
git log --oneline --graph --all
git push
```



```

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices
tyle)
$ nano css/style.css

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices
tyle)
$ git commit -m "[CSS] change bg to green"
On branch style
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working director
      modified:   css/style.css

no changes added to commit (use "git add" and/or "git commit -a")

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices
tyle)
$ git add css/style.css

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices
tyle)
$ git commit -m "[CSS] change bg to green"
[style ff55261] [CSS] change bg to green
1 file changed, 1 insertion(+), 1 deletion(-)

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices
tyle)
$ git checkout master
Switched to branch 'master'

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices
aster)
$ cat css/style.css
body {
  background: red;
}

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices
aster)
$ |

```

Fig. 22. – Capture 22 - Debut de la phase de finalisation.

```
MINGW64:/d/ingenierie-et-maquette-numerique/git-exercices
21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices
aster)
$ nano html/index.html

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices
aster)
$ git status
On branch master
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working director
      modified:   html/index.html

no changes added to commit (use "git add" and/or "git commit -a")

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices
aster)
$ git add html/index.html
warning: in the working copy of 'html/index.html', CRLF will be repla
d by LF the next time Git touches it

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices
aster)
$ git commit -m "[HTML] added h1 in index"
[master e567c1d] [HTML] added h1 in index
1 file changed, 2 insertions(+), 1 deletion(-)

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices
aster)
$ git checkout style
Switched to branch 'style'

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices
tyle)
$ cat css/style.css
body {
  background: green;
}

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exercices
tyle)
$
```

Fig. 23. – Capture 23 - Verification avant fusion ou publication.

```
MINGW64:/d/ingenierie-et-maquette-numerique/git-exe...
21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exerc
ices (style)
$ git checkout master
Switched to branch 'master'

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exerc
ices (master)
$ git merge style
Merge made by the 'ort' strategy.
css/style.css | 3 ++-
1 file changed, 2 insertions(+), 1 deletion(-)

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exerc
ices (master)
$ cat css/style.css
body {
  background: green;
  font-size: 24px;
}

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exerc
ices (master)
$ cat html/index.html
</!DOCTYPE html>
<html class="no-js">
  <head>
    <meta charset="utf-8">
    <meta http-equiv="X-UA-Compatible" content="IE=edge">
    <title>Git exercices</title>
    <meta name="description" content="">
    <meta name="viewport" content="width=device-width, initial-scal
e=1">
    <link rel="stylesheet" href="../css/style.css">
  </head>
  <body>
    <h1> ajouter une ligne</h1>
    <script src="../js/index.js" async defer></script>
  </body>
</html>

21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/git-exerc
ices (master)
$
```

Fig. 24. – Capture 24 - Operation de fusion ou de controle.



```
MINGW64:/d/ingenierie-et-maquette-numerique/new-repo
21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/new-repo (branch-a)
$ git status
On branch branch-a

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        conflit.txt

nothing added to commit but untracked files present (use "git add" to track)
21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/new-repo (branch-a)
$ git add *
21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/new-repo (branch-a)
$ git commit "change line 2"
error: pathspec 'change line 2' did not match any file(s) known to git
21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/new-repo (branch-a)
$ git commit -m "change line 2"
[branch-a (root-commit) 4f93fac] change line 2
1 file changed, 4 insertions(+)
create mode 100644 conflit.txt
21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/new-repo (branch-a)
$ |
```

Fig. 27. – Capture 27 - Verification de la branche courante.

```
MINGW64:/d/ingenierie-et-maquette-numerique/new-repo
21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/new-repo (master)
$ git branch
  branch-a
  branch-b
* master
21260@AHMED MINGW64 /d/ingenierie-et-maquette-numerique/new-repo (master)
$ |
```

Fig. 28. – Capture 28 - Derniere capture du TP.

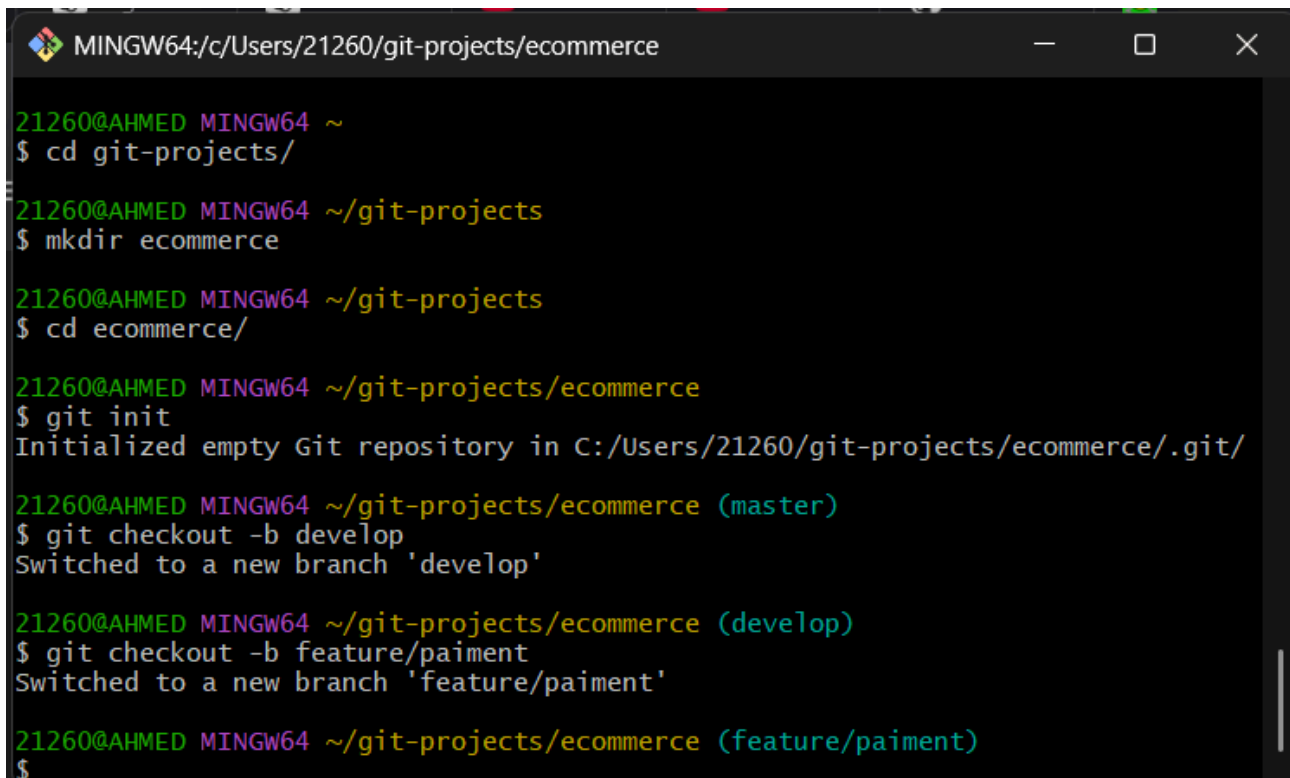
## Scenario complémentaire : depot e-commerce

---

### Objectif

Pour ancrer ces notions dans un cas realiste, un second scenario reproduit l'organisation d'un projet applicatif. Un depot ecommerce est cree, puis le travail se structure autour d'une branche develop pour l'integration et d'une branche feature/paiement dediee a une fonctionnalite. Ce decoupage prepare la fusion finale, qui revelera un conflit a resoudre.

```
mkdir ecommerce
cd ecommerce
git init
git checkout -b develop
git checkout -b feature/paiement
```



```
MINGW64:/c/Users/21260/git-projects/ecommerce

21260@AHMED MINGW64 ~
$ cd git-projects/

21260@AHMED MINGW64 ~/git-projects
$ mkdir ecommerce

21260@AHMED MINGW64 ~/git-projects
$ cd ecommerce/

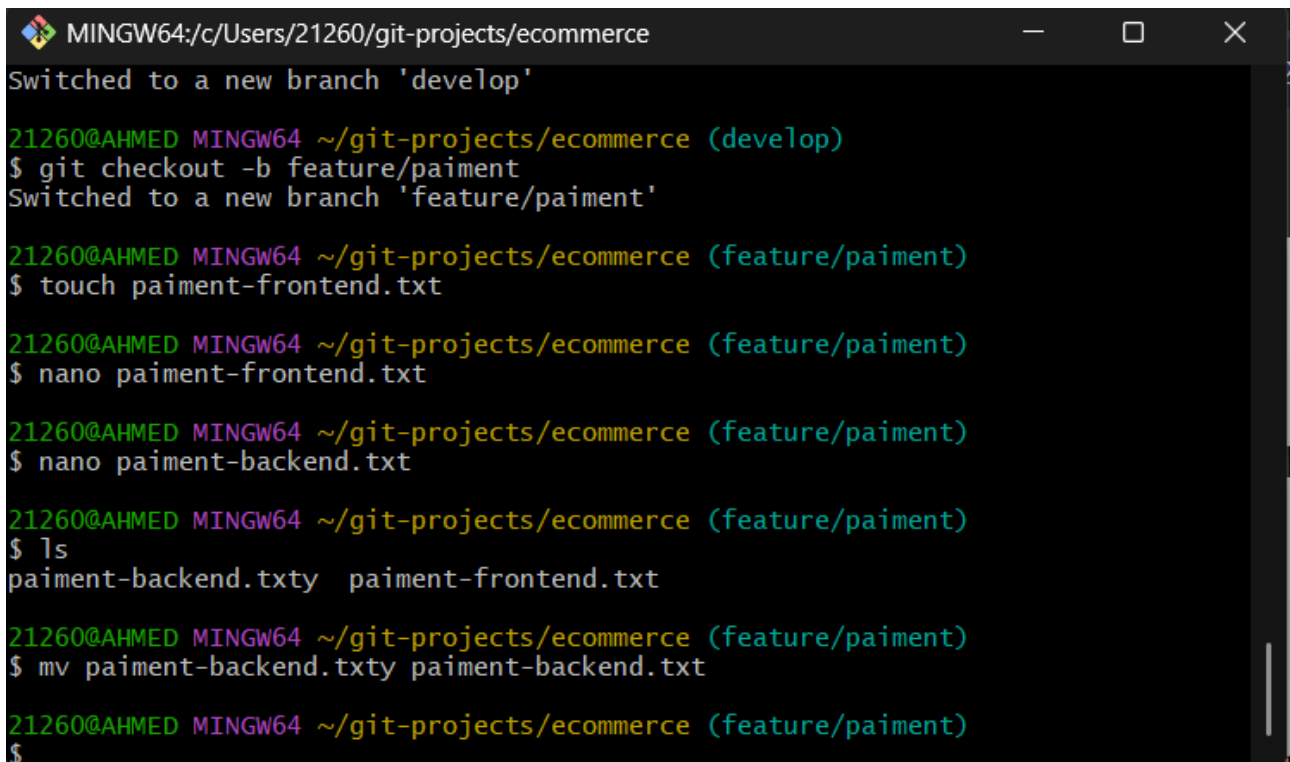
21260@AHMED MINGW64 ~/git-projects/ecommerce
$ git init
Initialized empty Git repository in C:/Users/21260/git-projects/ecommerce/.git/

21260@AHMED MINGW64 ~/git-projects/ecommerce (master)
$ git checkout -b develop
Switched to a new branch 'develop'

21260@AHMED MINGW64 ~/git-projects/ecommerce (develop)
$ git checkout -b feature/paiement
Switched to a new branch 'feature/paiement'

21260@AHMED MINGW64 ~/git-projects/ecommerce (feature/paiement)
$
```

Fig. 29. – Capture 29 - Initialisation du depot e-commerce et creation des branches.

A terminal window titled 'MINGW64:/c/Users/21260/git-projects/ecommerce' with standard Windows window controls. The terminal shows a sequence of Git commands and their outputs. It starts with switching to the 'develop' branch, then creating a new branch 'feature/paiement'. The user creates two files, 'paiement-frontend.txt' and 'paiement-backend.txt', using 'touch' and 'nano' respectively. A listing command 'ls' shows both files. Finally, the user renames 'paiement-backend.txt' to 'paiement-backend.txt' (a typo in the original image).

```
MINGW64:/c/Users/21260/git-projects/ecommerce
Switched to a new branch 'develop'

21260@AHMED MINGW64 ~/git-projects/ecommerce (develop)
$ git checkout -b feature/paiement
Switched to a new branch 'feature/paiement'

21260@AHMED MINGW64 ~/git-projects/ecommerce (feature/paiement)
$ touch paiement-frontend.txt

21260@AHMED MINGW64 ~/git-projects/ecommerce (feature/paiement)
$ nano paiement-frontend.txt

21260@AHMED MINGW64 ~/git-projects/ecommerce (feature/paiement)
$ nano paiement-backend.txt

21260@AHMED MINGW64 ~/git-projects/ecommerce (feature/paiement)
$ ls
paiement-backend.txt  paiement-frontend.txt

21260@AHMED MINGW64 ~/git-projects/ecommerce (feature/paiement)
$ mv paiement-backend.txt payment-backend.txt

21260@AHMED MINGW64 ~/git-projects/ecommerce (feature/paiement)
$
```

Fig. 30. – Capture 30 - Travail dans la branche feature/paiement.

## Publication et retour sur develop

La publication vers un depot distant suppose qu'un origin ait ete declare au prealable ; sans cela, `git push` echoue, ce qui rappelle la difference entre depot local et depot distant. De meme, revenir sur `develop` n'est possible que si la branche existe localement, d'où sa creation explicite.

```
git push origin feature/paiement
git checkout develop
git checkout -b develop
```

```
MINGW64:/c/Users/21260/git-projects/ecommerce
21260@AHMED MINGW64 ~/git-projects/ecommerce (feature/paiment)
$ git push origin feature/paiment
fatal: 'origin' does not appear to be a git repository
fatal: Could not read from remote repository.

Please make sure you have the correct access rights
and the repository exists.

21260@AHMED MINGW64 ~/git-projects/ecommerce (feature/paiment)
$ git checkout
21260@AHMED MINGW64 ~/git-projects/ecommerce (feature/paiment)
$ git checkout develop
error: pathspec 'develop' did not match any file(s) known to git

21260@AHMED MINGW64 ~/git-projects/ecommerce (feature/paiment)
$ git checkout develop
error: pathspec 'develop' did not match any file(s) known to git

21260@AHMED MINGW64 ~/git-projects/ecommerce (feature/paiment)
$ git checkout -b develop
Switched to a new branch 'develop'

21260@AHMED MINGW64 ~/git-projects/ecommerce (develop)
$ git merge feature/paiment
merge: feature/paiment - not something we can merge

21260@AHMED MINGW64 ~/git-projects/ecommerce (develop)
$ git merge feature/paiment develop
Already up to date.

21260@AHMED MINGW64 ~/git-projects/ecommerce (develop)
$
21260@AHMED MINGW64 ~/git-projects/ecommerce (develop)
$ ls
paiement-backend.txt  paiement-frontend.txt

21260@AHMED MINGW64 ~/git-projects/ecommerce (develop)
$
```

Fig. 31. – Capture 31 - Erreur de depot distant et creation de la branche develop.

## Fusion et verification des fichiers

La branche de fonctionnalite est ensuite integree dans develop. Une fois la fusion realisee, on verifie que les fichiers produits pendant le scenario sont bien presents et que leur contenu correspond a ce qui etait attendu.

```
git merge feature/paiment
ls
cat paiement-frontend.txt
cat paiement-backend.txt
```



```
MINGW64:/c/Users/21260/git-projects/ecommerce
$ git merge feature/paiement
merge: feature/paiement - not something we can merge

1260@AHMED MINGW64 ~/git-projects/ecommerce (develop)
$ git merge feature/paiement develop
Already up to date.

1260@AHMED MINGW64 ~/git-projects/ecommerce (develop)
$
1260@AHMED MINGW64 ~/git-projects/ecommerce (develop)
$ ls
paiement-backend.txt  paiement-frontend.txt

1260@AHMED MINGW64 ~/git-projects/ecommerce (develop)
$ git checkout dev1
error: pathspec 'dev1' did not match any file(s) known to git

1260@AHMED MINGW64 ~/git-projects/ecommerce (develop)
$ git checkout -b dev1
switched to a new branch 'dev1'

1260@AHMED MINGW64 ~/git-projects/ecommerce (dev1)
$ nano
git/      paiement-backend.txt  paiement-frontend.txt

1260@AHMED MINGW64 ~/git-projects/ecommerce (dev1)
$ nano paiement-frontend.txt

1260@AHMED MINGW64 ~/git-projects/ecommerce (dev1)
$ nano paiement-frontend.txt

1260@AHMED MINGW64 ~/git-projects/ecommerce (dev1)
$ cat paiement-frontend.txt
code frontend du paiement

code developer 1

1260@AHMED MINGW64 ~/git-projects/ecommerce (dev1)
$
```

Fig. 32. – Capture 32 - Fusion et verification des fichiers de paiement.

```
MINGW64:/c/Users/21260/git-projects/ecommerce
ode developer 1

1260@AHMED MINGW64 ~/git-projects/ecommerce (dev1)
$ git checkout -b dev2
Switched to a new branch 'dev2'

1260@AHMED MINGW64 ~/git-projects/ecommerce (dev2)
$ nano paiement-frontend.txt

1260@AHMED MINGW64 ~/git-projects/ecommerce (dev2)
$ git checkout -b dev1
fatal: a branch named 'dev1' already exists

1260@AHMED MINGW64 ~/git-projects/ecommerce (dev2)
$ git checkout dev1
Switched to branch 'dev1'
$ nano paiement-frontend.txt

1260@AHMED MINGW64 ~/git-projects/ecommerce (dev1)
$ git add .

1260@AHMED MINGW64 ~/git-projects/ecommerce (dev1)
$ git commit -m "code developer 1"
[dev1 13625bf] code developer 1
1 file changed, 2 insertions(+)

1260@AHMED MINGW64 ~/git-projects/ecommerce (dev1)
$ git checkout dev
error: pathspec 'dev' did not match any file(s) known to git

1260@AHMED MINGW64 ~/git-projects/ecommerce (dev1)
$ git checkout develop
Switched to branch 'develop'

1260@AHMED MINGW64 ~/git-projects/ecommerce (develop)
$ git branch -m dev2
fatal: a branch named 'dev2' already exists

1260@AHMED MINGW64 ~/git-projects/ecommerce (develop)
$
```

Fig. 33. – Capture 33 - Consultation du contenu des fichiers.

## Conflit et resolution

Lorsque deux branches modifient la meme zone d'un fichier, Git ne peut pas trancher seul : il signale un conflit sur paiement-frontend.txt et suspend la validation. La resolution est manuelle : on edite le fichier pour choisir la version correcte, puis on confirme avec `git add` et `git commit`, ce qui debloque l'historique.

```
git merge <branche>
nano paiement-frontend.txt
git add .
git commit -m "resolve merge conflicts"
```

```
MINGW64:/c/Users/21260/git-projects/ecommerce
[develop 1bbc33c] code developoer 2
1 file changed, 2 insertions(+)

21260@AHMED MINGW64 ~/git-projects/ecommerce (develop)
$ cat paiement-frontend.txt
code frontend du paiement

code developoer 2

21260@AHMED MINGW64 ~/git-projects/ecommerce (develop)
$ git merge dev1 developoer
merge: developoer - not something we can merge

21260@AHMED MINGW64 ~/git-projects/ecommerce (develop)
$ git merge dev1
Auto-merging paiement-frontend.txt
CONFLICT (content): Merge conflict in paiement-frontend.txt
Automatic merge failed; fix conflicts and then commit the result.

21260@AHMED MINGW64 ~/git-projects/ecommerce (develop|MERGING)
$ ls
paiement-backend.txt  paiement-frontend.txt

21260@AHMED MINGW64 ~/git-projects/ecommerce (develop|MERGING)
$ git paiement-frontend.txt
git: 'paiement-frontend.txt' is not a git command. See 'git --help'.

21260@AHMED MINGW64 ~/git-projects/ecommerce (develop|MERGING)
$ nano paiement-frontend.txt

21260@AHMED MINGW64 ~/git-projects/ecommerce (develop|MERGING)
$ cat paiement-frontend.txt
code frontend du paiement

code developoer 2
code developoer 1

21260@AHMED MINGW64 ~/git-projects/ecommerce (develop|MERGING)
$
```

Fig. 34. – Capture 34 - Detection du conflit pendant la fusion.

```
MINGW64:/c:/Users/21260/git-projects/ecommerce
Auto-merging paiement-frontend.txt
CONFLICT (content): Merge conflict in paiement-frontend.txt
Automatic merge failed; fix conflicts and then commit the result.

21260@AHMED MINGW64 ~/git-projects/ecommerce (develop|MERGING)
$ ls
paiement-backend.txt  paiement-frontend.txt

21260@AHMED MINGW64 ~/git-projects/ecommerce (develop|MERGING)
$ git paiement-frontend.txt
git: 'paiement-frontend.txt' is not a git command. See 'git --help'.

21260@AHMED MINGW64 ~/git-projects/ecommerce (develop|MERGING)
$ nano paiement-frontend.txt

21260@AHMED MINGW64 ~/git-projects/ecommerce (develop|MERGING)
$ cat paiement-frontend.txt
code frontend du paiement

code developer 2
code developer 1

21260@AHMED MINGW64 ~/git-projects/ecommerce (develop|MERGING)
$ git commit -m "resolve merge conflicts"
error: Committing is not possible because you have unmerged files.
hint: Fix them up in the work tree, and then use 'git add/rm <file>'
hint: as appropriate to mark resolution and make a commit.
fatal: Exiting because of an unresolved conflict.
U      paiement-frontend.txt

21260@AHMED MINGW64 ~/git-projects/ecommerce (develop|MERGING)
$ git add .

21260@AHMED MINGW64 ~/git-projects/ecommerce (develop|MERGING)
$ git commit -m "resolve merge conflicts"
[develop 2e9d4c5] resolve merge conflicts

21260@AHMED MINGW64 ~/git-projects/ecommerce (develop)
$ |
```

Fig. 35. – Capture 35 - Resolution et commit final du conflit.

## Conclusion

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Ce TP a permis de parcourir le cycle de vie complet d'un projet sous Git : initialiser un depot, suivre les fichiers, enregistrer des commits, travailler en parallele grace aux branches, comparer les versions, fusionner et resoudre un conflit. Chaque etape s'appuie sur une trace verifiable, ce qui rend le travail reproductible.

Maitriser ces commandes constitue la base du travail collaboratif sur GitHub : tout changement devient identifiable, discutable et integrable proprement, ce qui donne a une equipe un historique clair et une plus grande confiance dans le code partage.